

2 Facetted Graphs Guided Lab

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9/16/25

No Facets (Yet)

```
require(reshape2)
```

```
## Loading required package: reshape2
```

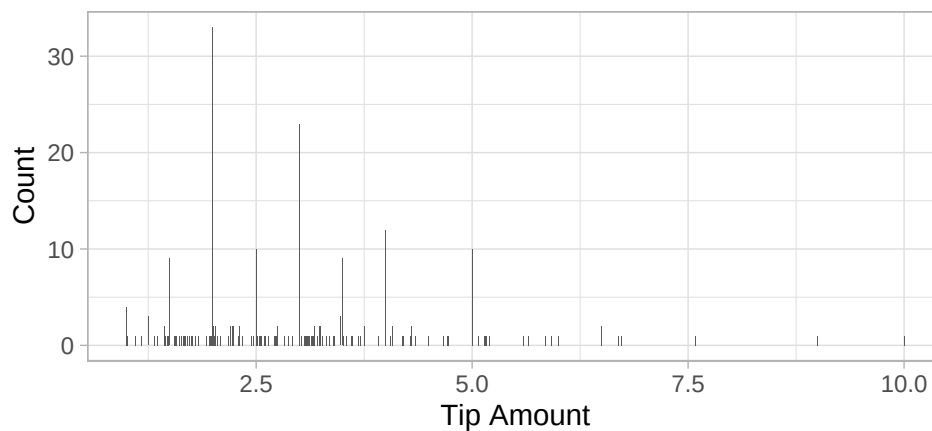
```
head(tips)
```

```
##   total_bill  tip  sex smoker day  time size
## 1    16.99  1.01 Female    No  Sun  Dinner    2
## 2    10.34  1.66   Male    No  Sun  Dinner    3
## 3    21.01  3.50   Male    No  Sun  Dinner    3
## 4    23.68  3.31   Male    No  Sun  Dinner    2
## 5    24.59  3.61 Female    No  Sun  Dinner    4
## 6    25.29  4.71   Male    No  Sun  Dinner    4
```

```
gf_bar( ~ tip , data = tips)%>%gf_labs(x = "Tip Amount", y = "Count", title="Exploring Tips" , subtitle
```

Exploring Tips

Bar Chart

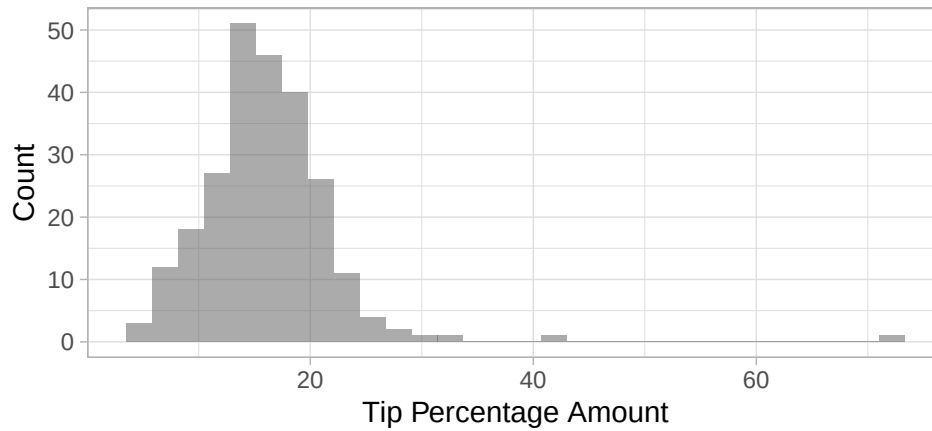


```
gf_histogram(~ tip / total_bill * 100 , data = tips)%>%gf_labs(x = "Tip Percentage Amount", y = "Count"
```

```
## `stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

Exploring Tip Percentages

Bar Chart

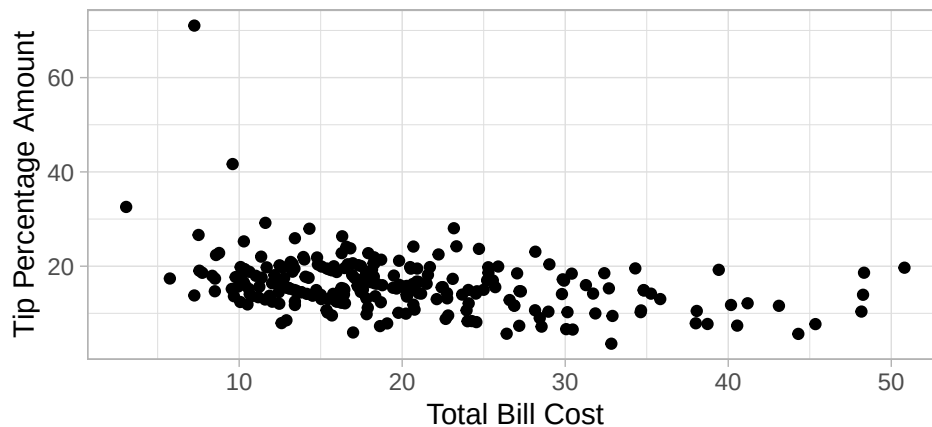


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart



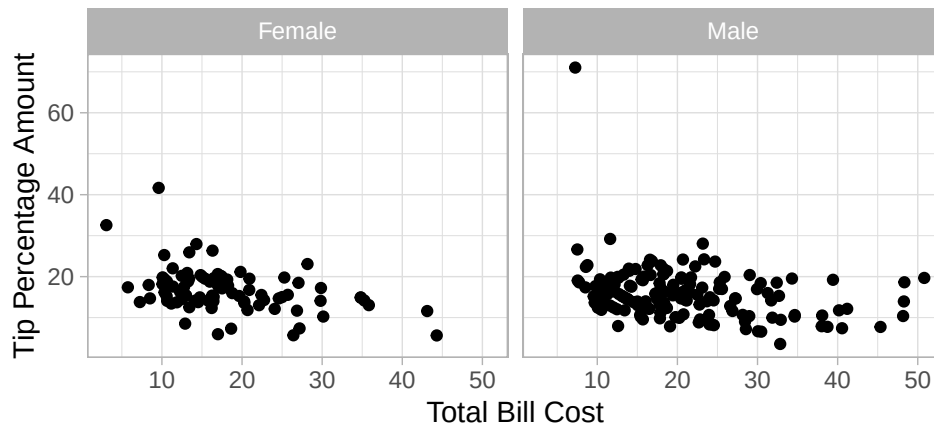
Dataset: tips(reshape2)

Facet By Sex

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrapped(Sex)

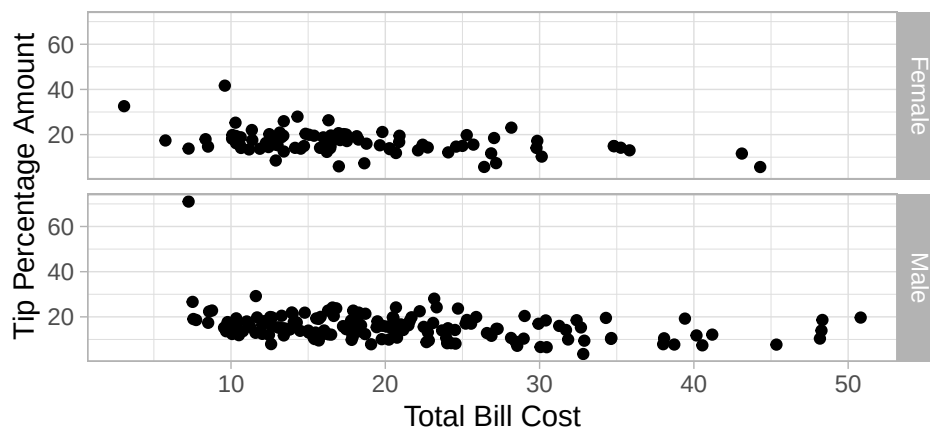


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrapped(Sex Adjusted)

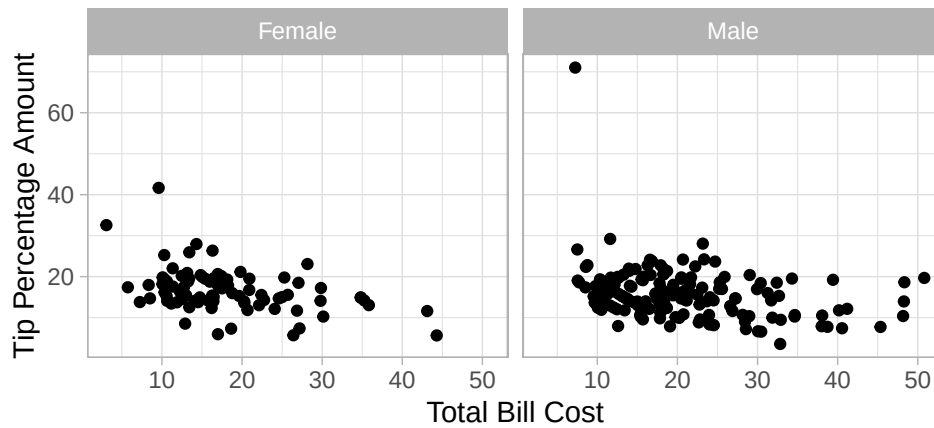


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Grid(Sex X)

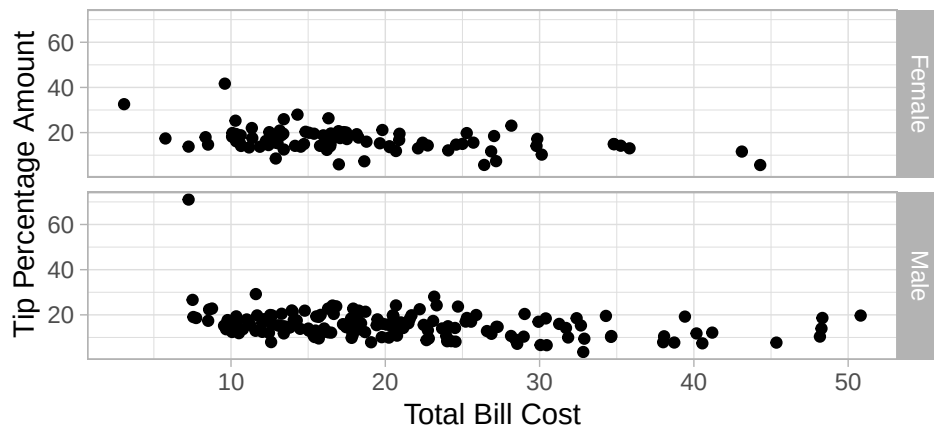


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Grid(Sex Y)



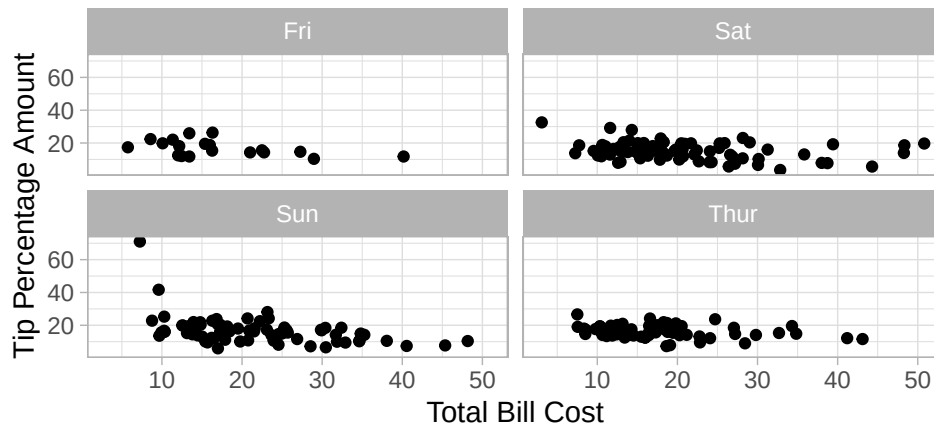
Dataset: tips(reshape2)

Facet by Day

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrapped(day)



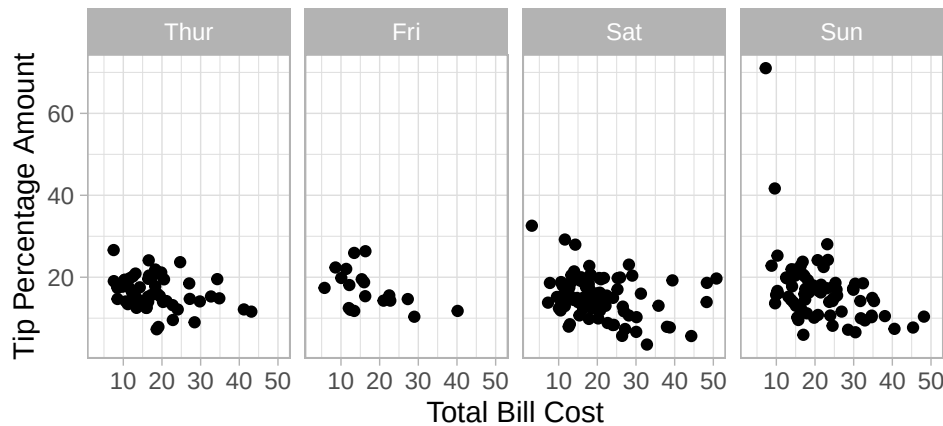
Dataset: tips(reshape2)

```
tips$day=factor(tips$day,levels=c("Thur","Fri","Sat","Sun"))
```

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrapped(day) – nrow=1

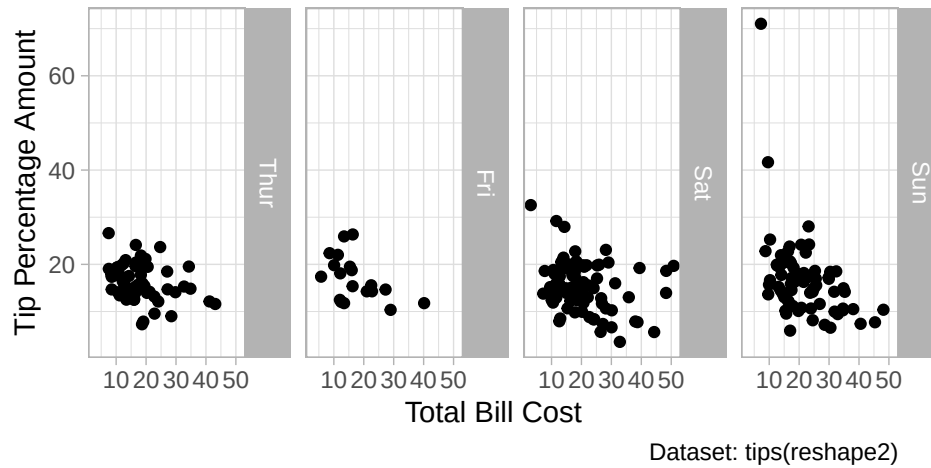


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrapped(day) – Strip Right

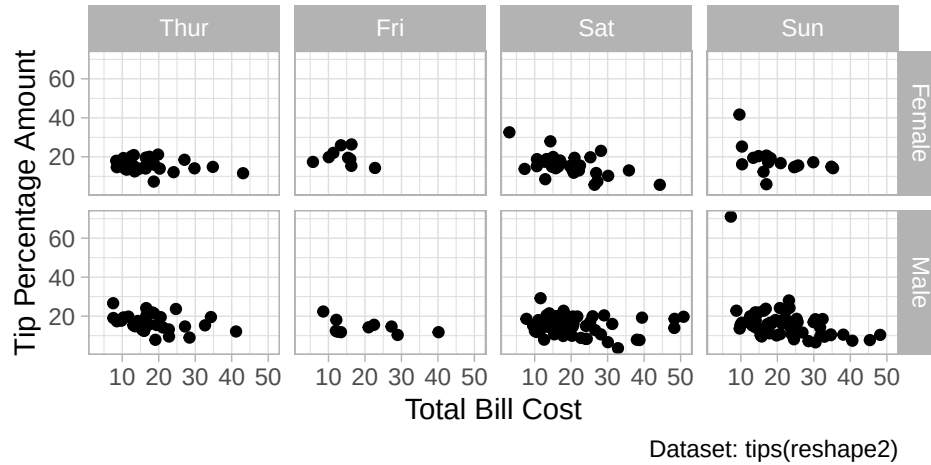


Facet by BOTH Sex & Day

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

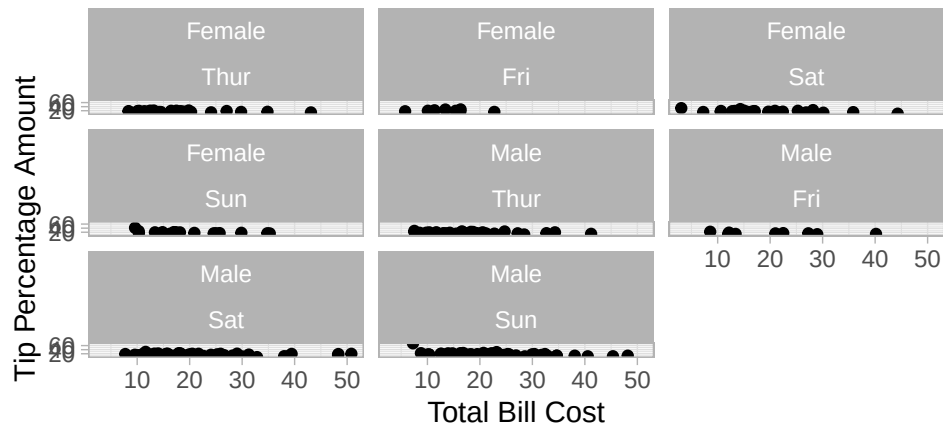
Bar Chart – Facet Grid(Day, Sex)



```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrap(Day, Sex)

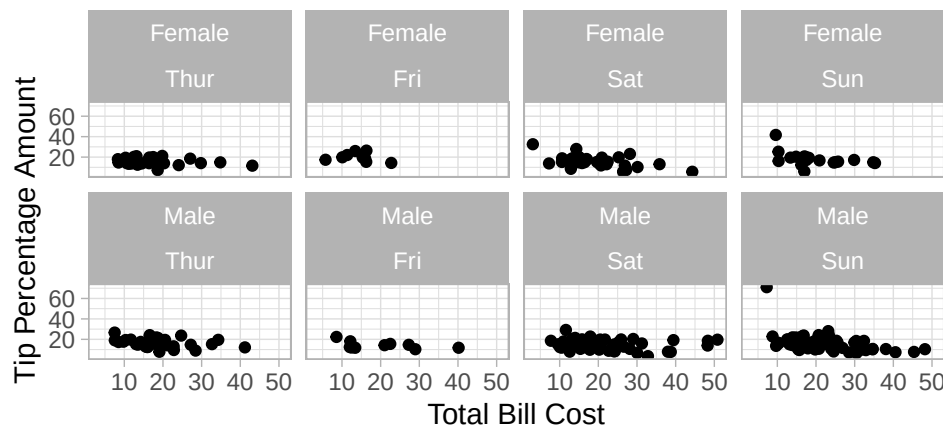


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Wrap(Day, Sex)



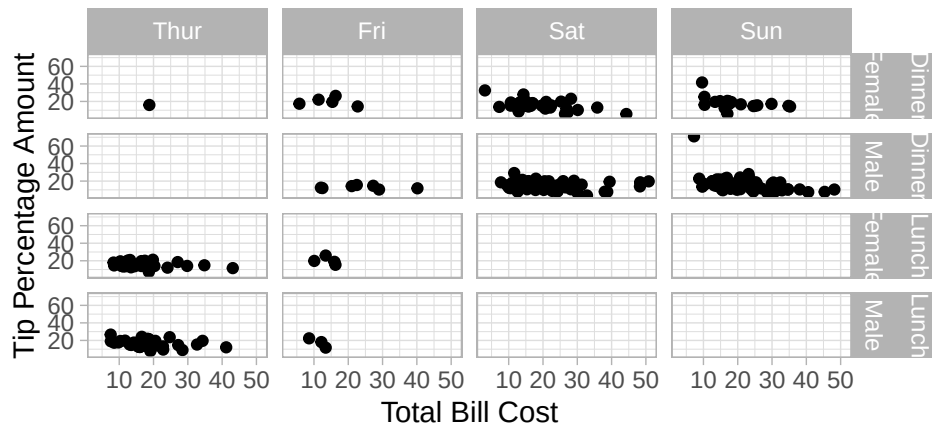
Dataset: tips(reshape2)

Facet with Three Variables

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

Bar Chart – Facet Grid(Day, Time+Sex)

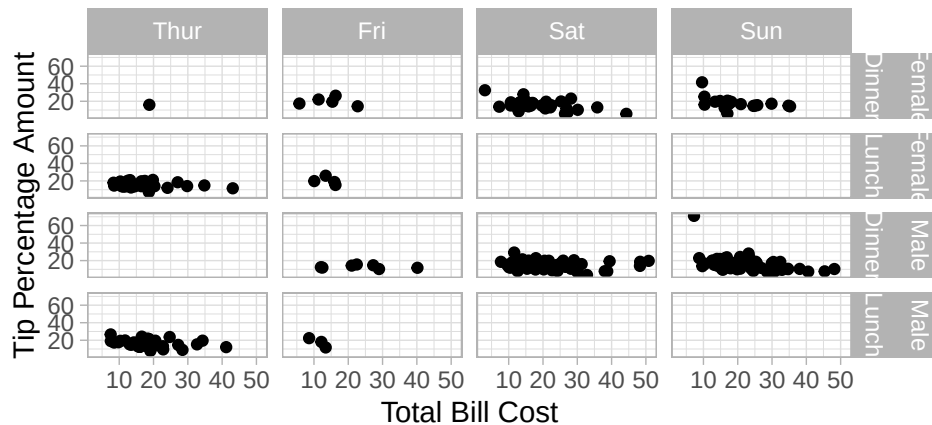


Dataset: tips(reshape2)

```
gf_point( tip*100/total_bill ~ total_bill , data = tips )>%gf_labs(x = "Total Bill Cost", y = "Tip Per
```

Tip Percentages explained by Bill Cost

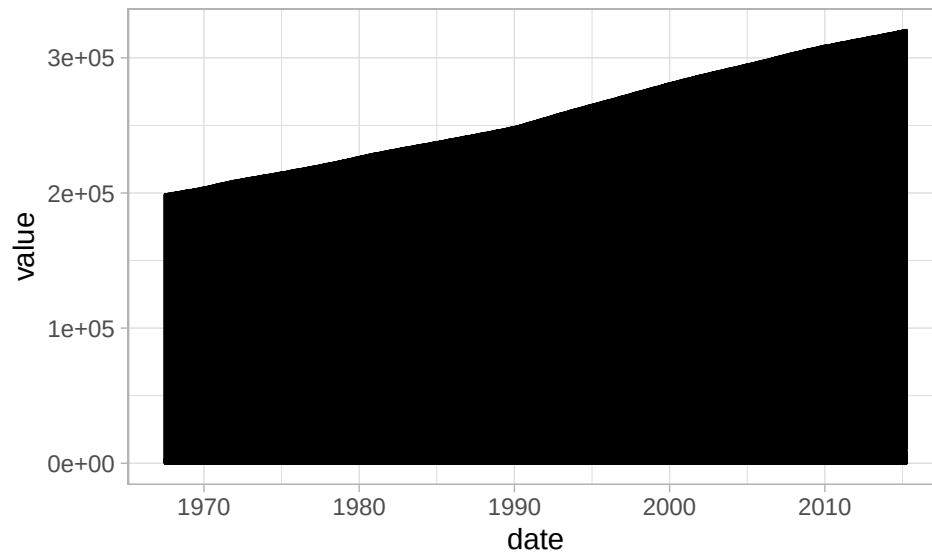
Bar Chart – Facet Grid(Day, Sex+Time)



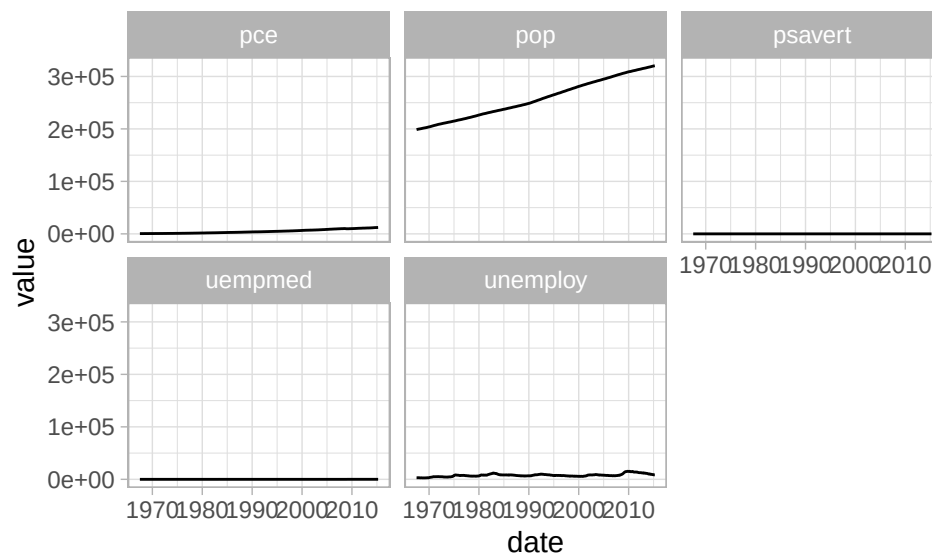
Dataset: tips(reshape2)

Free Scales

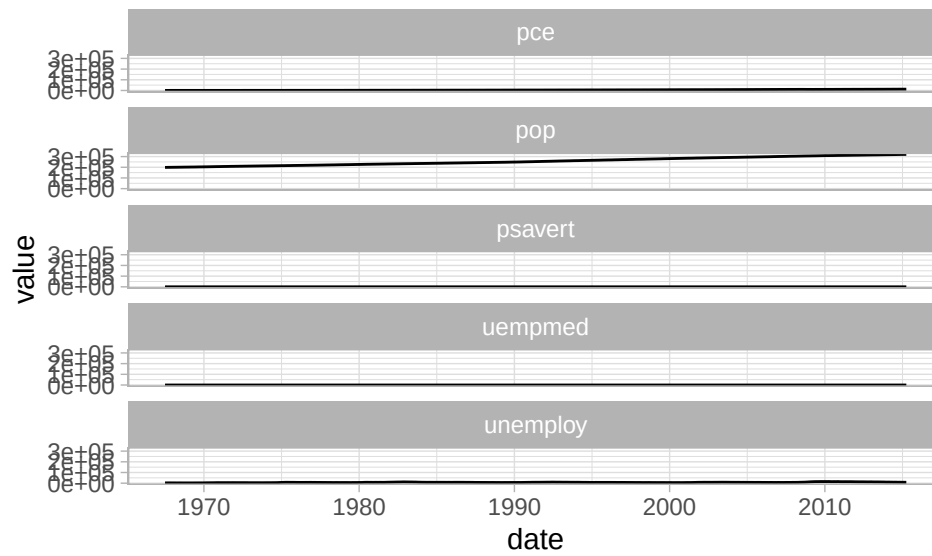
```
gf_line( value ~ date , data = economics_long )
```



```
gf_line( value ~ date , data = economics_long )>% gf_facet_wrap(~variable)
```



```
gf_line( value ~ date , data = economics_long )>% gf_facet_wrap(~variable,ncol=1)
```



```
gf_line( value ~ date , data = economics_long ) %>% gf_facet_wrap(~variable, ncol=1, scales = "free_y")
```

